

AUTOMATIC FARE COLLECTION SYSTEM WITH VOICE ALERT

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ಬೆಂಗಳೂರು ನಗರ ವಿಶ್ವವಿದ್ಯಾನಿಲಯ



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Submitted for fulfilment of the requirement of Mini project (3MCA7) for

MASTER OF COMPUTER APPLICATIONS

Submitted by

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Under the Guidance of

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CERTIFICATE

This is to certify that the report entitled “**Automatic Fare Collection System With Voice Alert**” embodies the original work by **GAGANA C P (P18CD24S126020)** the fulfilment of the requirements for Mini Project for MCA, III Semester course during the academic semester from January 2026 to March 2026 as prescribed by Dr. Manmohan Singh Bengaluru City University.

Project Guide

Head of the Department

(Mrs. ASHWINI D N)

(Dr. MANJULA P S)

Place: Bangalore

Date:

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Project Guide

(Mrs. ASHWINI D N)

Head of the Department

(Dr. MANJULA P S)

Place: Bangalore

Date:

External Examiner

1. _____

2. _____

DECLARATION

I, **GAGANA C P** do hereby declare that the project work entitled “**Automatic Fare Collection System With Voice Alert**” is a Bonafide work carried out by us under the guidance of **Mrs. ASHWINI D N**, this project as presented in this report is our original work and has not been presented for any other University award. This project has been submitted as partial fulfilment of requirement for the Degree of Master of Computer Applications 3rd Semester of Dr. Manmohan Singh Bengaluru City University.

GAGANA C P

Acknowledgement

I would like to express our gratitude and appreciation to all those who gave us the possibility to complete this report.

We take pleasure in expressing gratitude towards our institution for providing us the opportunity to work on this project as a part of our course. We would like to express our gratitude and sincere thanks to our management, we would like to express our gratitude to Principal & Director **Dr. Janardhan T.G** for providing the right atmosphere at the institute. We also wish to express our heart full thanks to **Dr. Manjula P S**, Head of the Department, for being helpful through the project.

We express our deep sense of gratitude to our project guide

Mrs. ASHWINI D N, Associate professor, for her unfailing encouragement, valuable guidance and suggestions to us in the course of project work. we also express our heartfelt thanks to our parents for being supportive in all our activities without whom it wouldn't have been possible for us to complete our project.

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SYNOPSIS

1.SYNOPSIS

Aim

The aim of this project is to design and implement an Automatic Fare Collection System with Voice Alert for public bus transportation using Arduino UNO and RFID technology. This system helps to automate ticket collection, reduce manual errors, prevent fare evasion, and provide a smooth, secure, and contactless travel experience for passengers. It automatically identifies passengers using RFID cards, calculates fare based on the number of stops travelled, deducts the balance, controls entry and exit gates using servo motors, and gives voice announcements of bus stops.

Description

The Automatic Fare Collection System with Voice Alert is an IoT-based smart bus ticketing system developed using Arduino UNO, RFID Reader, RFID cards, 16x2 I2C LCD display, DFPlayer Mini MP3 module, speaker, servo motors, DC motor, and L293D motor driver. The system is designed to improve public transportation by making fare collection faster, easier, and more efficient.

When the START push button is pressed, the DC motor starts and represents bus movement. As the bus reaches each stop, the STOP button is pressed to stop the bus, and the DFPlayer Mini plays a pre-recorded voice announcement of the stop name through the speaker.

Passengers enter the bus by scanning their RFID cards on the MFRC522 RFID Reader. The system verifies the card, checks the available balance, records the boarding stop, and opens the entry gate using a servo motor. When the passenger reaches the destination, the RFID card is scanned again. The system calculates the number of stops travelled, deducts the fare at Rs.10 per stop, and opens the exit gate.

All important details such as passenger name, boarding stop, destination stop, fare amount, and remaining balance are displayed on the LCD screen. The route covers 10 bus stops from Kempe Gowda Bus Stand to 16th Main BTM, making this project a practical model for smart and automated public transport systems

INTRODUCTION

The Automatic Fare Collection System with Voice Alert is an Arduino UNO based smart transportation project developed to improve the traditional bus ticketing system. In many public buses, ticket collection is done manually, which takes more time and may cause errors in fare calculation. This project uses RFID technology to automate fare collection and make the process faster, safer, and more efficient.

Passengers use RFID cards for entry and exit, and the system automatically calculates the fare based on the number of stops travelled. The LCD display shows passenger details, balance, and stop information, while the DFPlayer Mini provides voice alerts for stop announcements and card status. Servo motors control the entry and exit gates, and a DC motor is used to simulate bus movement. This system helps reduce manual work, improves passenger convenience, and supports smart public transport management.

MODULES

Module 1: System Initialization and Bus Control

This module initializes all hardware components such as Arduino UNO, RFID Reader, LCD Display, DFPlayer Mini, Servo Motors, and DC Motor. When the START button is pressed, the DC motor starts and represents bus movement. When the STOP button is pressed, the motor stops, showing that the bus has reached a stop.

Module 2: RFID Card Scanning and Validation

This module reads the RFID card using the MFRC522 RFID Reader. When a passenger taps the card, the system checks the card UID with stored data. If the card is valid, the passenger is allowed to enter or exit. If the card is invalid, the LCD displays “Invalid Card” and access is denied.

Module 3: Fare Calculation and Balance Management

This module records the boarding stop when the passenger enters the bus. At the destination, the passenger scans the card again. The system calculates the fare based on the number of stops travelled at Rs.10 per stop and deducts the amount from the card balance.

If the balance is low, the system shows a warning message.

Module 4: Gate Control Using Servo Motors

Two servo motors are used for entry and exit gates. After successful card verification, the entry gate opens for boarding and closes automatically. Similarly, after fare deduction, the exit gate opens for the passenger to leave the bus.

Module 5: Voice Alert Announcement

The DFPlayer Mini MP3 module gives voice announcements for each bus stop. Audio files are stored in the SD card. When the bus reaches a stop, the speaker announces the stop name, helping passengers and improving travel convenience.

Module 6: LCD Display Output

The 16x2 I2C LCD Display shows important details such as passenger name, stop name, fare amount, remaining balance, and system messages like “Welcome”, “Tap Card”, “Invalid Card”, and “Low Balance”. It acts as the main display interface for the system.

CIRCUIT DIAGRAM

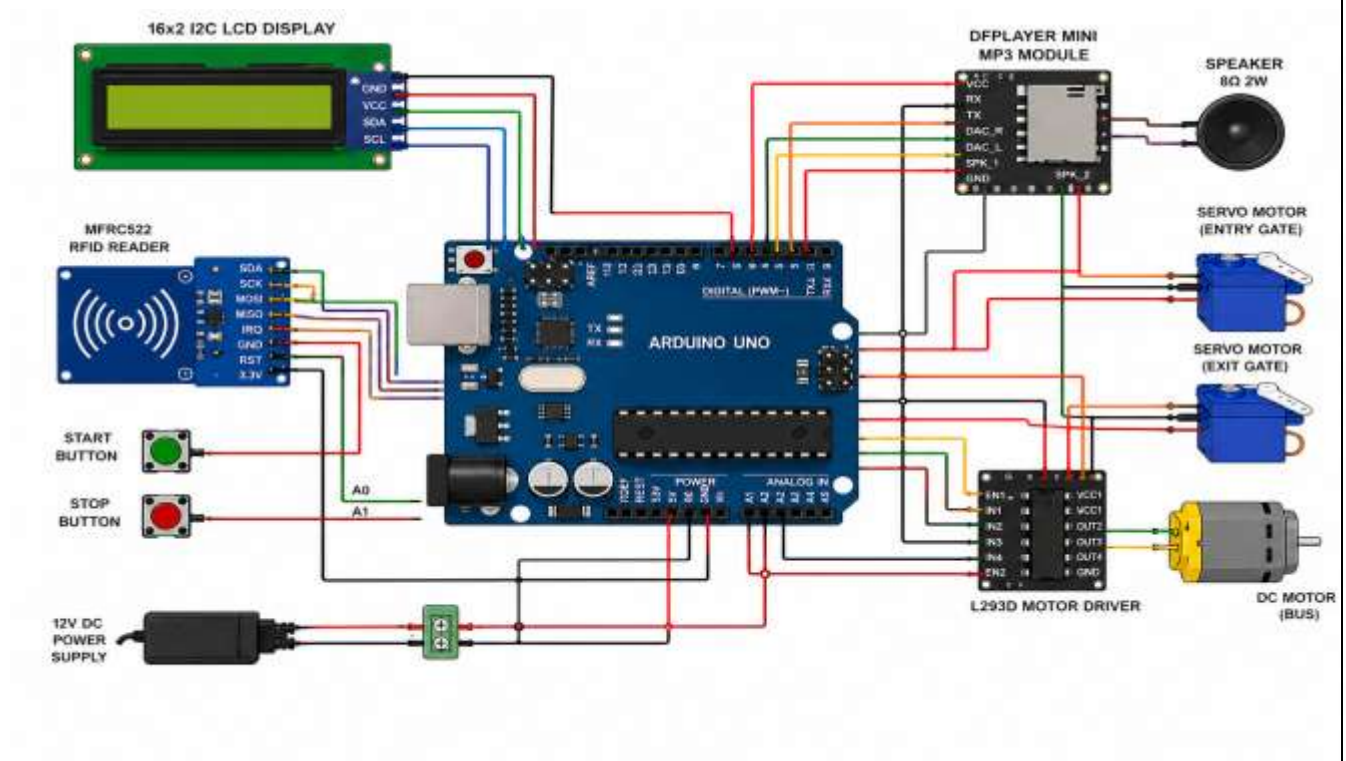


Figure 1: Circuit Diagram

The circuit diagram uses Arduino UNO as the main controller of the system. The MFRC522 RFID reader scans passenger cards for entry and exit, while the 16x2 I2C LCD display shows stop names, fare details, and card balance. The DFPlayer Mini with speaker provides voice alerts for stop announcements and card status.

Two servo motors are used to control the entry and exit gates automatically. The L293D motor driver controls the DC motor, which represents bus movement. START and STOP push buttons are used to operate the system, and a 12V power supply is used to run all the components efficiently.

DATA FLOW DIAGRAM (DFD)

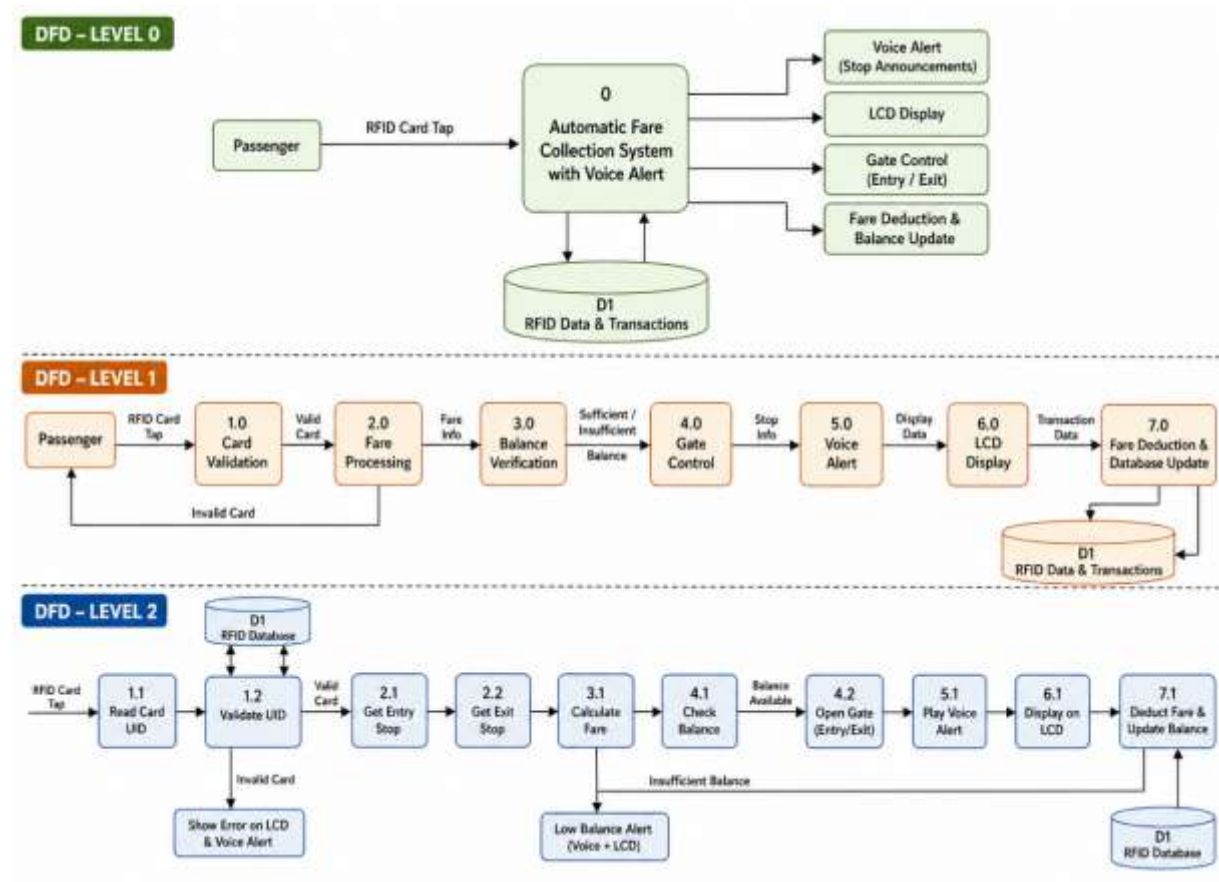


Figure 2: Data Flow Diagram

CONCLUSION

The Automatic Fare Collection System with Voice Alert provides a smart and efficient solution for public bus transportation. It reduces manual ticket collection by using RFID cards for automatic fare deduction based on travelled stops. The system also improves passenger convenience through LCD display updates, voice alerts, and automatic gate control.

This project helps in saving time, reducing human errors, and improving transport management. It is a low-cost and reliable system that can be further developed for real-time public transport applications in smart cities.

FUTURE SCOPE

The project opens avenues for further research and development. Future enhancements and extensions could include:

- The system can be implemented on a larger scale in public buses, metro buses, and school transportation systems for automatic fare collection and passenger management.
- GPS technology can be integrated to provide real-time bus tracking, live location updates, and accurate arrival time information for passengers.
- Online payment options such as UPI, mobile wallets, smart cards, and QR code-based payments can be added to support fully cashless transportation.
- A mobile application can be developed for passengers to check bus routes, current location, fare details, and receive notifications about bus arrival and delays.
- Cloud database integration can be used to store passenger details, travel history, and fare records securely for better transport management and report generation.
- The voice alert system can be improved by adding multi-language support so that passengers from different regions can easily understand stop announcements.
- Face recognition and biometric verification can be introduced for passenger identification and improved security in public transportation systems.
- AI-based passenger monitoring and crowd analysis can be implemented to improve route planning, reduce overcrowding, and enhance passenger safety.
- Automatic report generation for daily fare collection, passenger count, and route performance can help transport authorities in better decision-making.
- This project can be integrated into smart city transportation systems to support digital public transport and improve overall urban mobility management.

INTRODUCTION

2. INTRODUCTION

Public transportation plays an important role in daily life, especially in cities where many people depend on buses for travel. In many bus systems, ticket collection is still done manually by conductors who collect cash and issue tickets. This method often causes problems such as delays, human errors, fare evasion, and difficulty in managing change.

With the development of IoT (Internet of Things), embedded systems, and RFID technology, public transport systems can be made smarter and more efficient. RFID (Radio Frequency Identification) is a contactless technology used for identification and tracking. It is already used in metro cards, toll collection systems like FASTag, office access systems, and library management.

This project, “Automatic Fare Collection System with Voice Alert,” is designed using Arduino UNO and RFID technology to automate bus fare collection. The system uses an MFRC522 RFID Reader to scan passenger cards, a 16x2 I2C LCD to display information, servo motors for entry and exit gate control, a DFPlayer Mini MP3 module with speaker for stop announcements, and a DC motor with L293D motor driver to simulate bus movement.

When the bus starts, the DC motor runs using the START button. At each stop, the STOP button is pressed, and the speaker announces the stop name using pre-recorded voice messages. Passengers enter by scanning their RFID cards. The system checks the card balance, records the boarding stop, and opens the entry gate. At the destination, the passenger scans the card again, the fare is calculated at Rs.10 per stop, the amount is deducted, and the exit gate opens.

The route in this project covers 10 stops from Kempe Gowda Bus Stand to 16th Main BTM in Bengaluru. All passenger details such as name, stop, fare, and remaining balance are shown on the LCD display.

This project helps reduce manual work, improves fare management, prevents fraud, and provides a faster and safer travel experience. It is a simple and practical model for developing smart public transportation systems in the future.

LITERATURE REVIEW

3. LITERATURE REV LITERATURE REVIEW

1. RFID Based Automatic Bus Fare Collection System

Authors: R. Kumar, S. Prakash, and P. Sharma

Year: 2021

This paper explains the use of RFID technology for automatic bus fare collection. In traditional bus systems, conductors manually collect money and issue tickets, which consumes more time and may cause errors. The proposed RFID-based system uses smart cards for passenger identification and automatic fare deduction. When a passenger enters the bus, the RFID reader scans the card and stores the boarding details. On exit, the system calculates the fare based on the travel distance and deducts the amount automatically. This system reduces human effort, improves accuracy, and provides a faster travel experience for passengers.

2. Smart Bus Ticketing System Using Arduino UNO

Authors: M. Karthik, A. Rajesh, and V. Naveen

Year: 2020

This paper presents an Arduino UNO based smart ticketing system for public transportation. The system uses RFID cards, LCD display, and GSM technology for passenger management. Arduino UNO acts as the main controller and manages all operations like card validation, fare calculation, and display output. The LCD shows travel details and available balance. The research explains that using Arduino makes the system cost-effective, simple to implement, and suitable for real-time applications. This study supports the development of automatic fare systems for buses and metro services.

3. IoT Based Smart Public Transport Fare Collection System

Authors: S. Ramesh and K. Divya

Year: 2022

This research focuses on IoT-based fare collection systems for smart city transportation. The system combines RFID, cloud database, and GPS tracking for better transport management. Passenger details and fare records are stored in cloud storage for easy monitoring and report generation. GPS helps in live bus tracking and stop management. The paper highlights that IoT integration improves transparency, passenger convenience, and transport efficiency. This concept helps in developing future smart transportation systems with better automation.

4. Voice Announcement System for Smart Bus Transportation

Authors: P. Anitha and R. Suresh

Year: 2021

This paper explains the importance of voice alert systems in public buses. The system uses audio modules like DFPlayer Mini for stop announcements and passenger guidance. Voice alerts help passengers know the current stop, next stop, and important instructions. This is especially useful for elderly passengers and visually challenged people. The study shows that combining voice announcements with automatic fare collection improves user experience and makes the system more accessible and passenger-friendly.

5. RFID Based Passenger Entry and Exit Monitoring System

Authors: N. Harish, T. Kiran, and S. Tejas

Year: 2020

This paper discusses passenger monitoring using RFID technology for entry and exit control. The system records passenger movement and maintains count automatically. Servo motors are used for gate opening and closing after successful card verification. Invalid cards and low balance conditions are also detected by the system. The research explains that gate automation improves security and prevents unauthorized access. This concept is highly useful for bus fare automation projects.

6. Automatic Fare Calculation Using Embedded Systems

Authors: V. Srinivas and L. Deepak

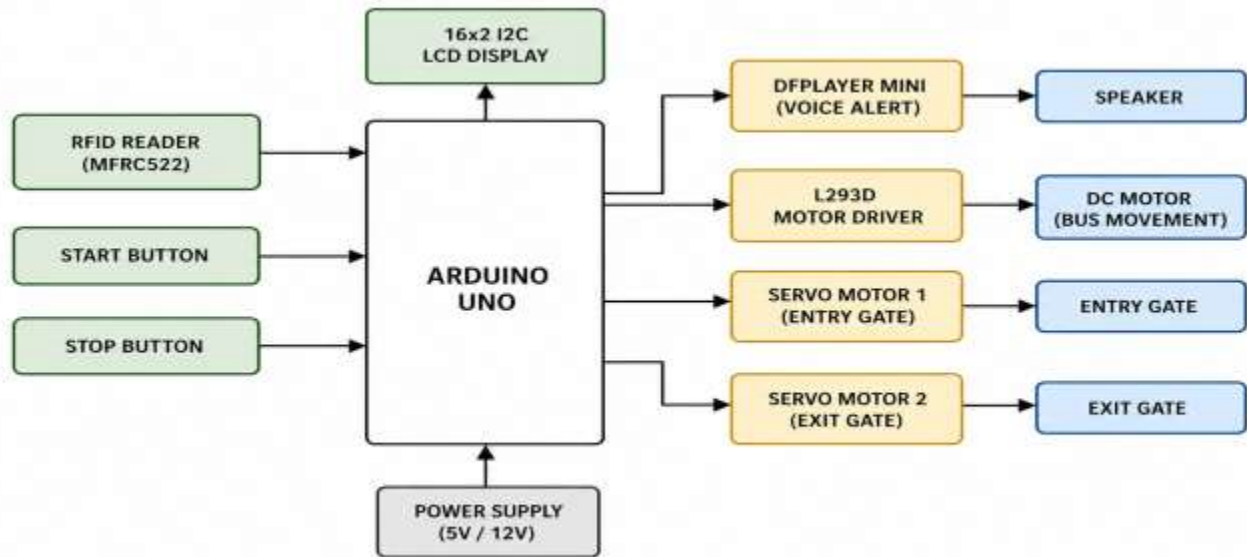
Year: 2019

This study focuses on automatic fare calculation using microcontrollers and embedded systems. Instead of fixed ticket prices, the system calculates fare based on the number of stops travelled by passengers. Boarding and destination stops are stored digitally, and the fare is deducted accordingly. This improves transparency and avoids manual mistakes. The paper proves that stop-based fare calculation is more accurate and fair compared to manual ticketing systems.

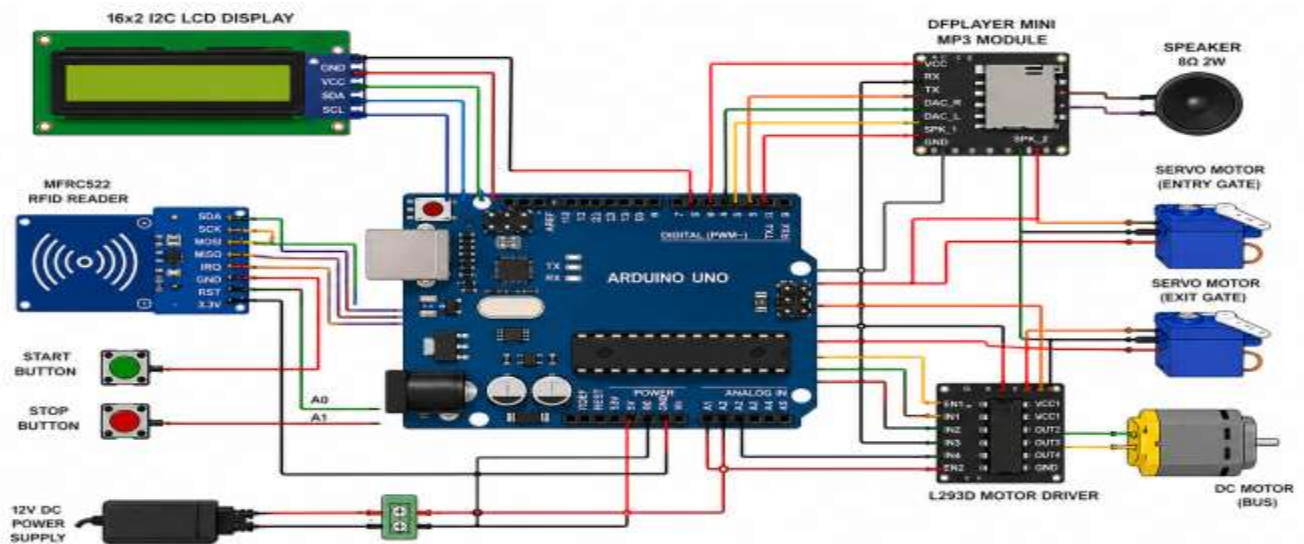
ARCHITECTURE

4. ARCHITECTURE

Block Diagram



Circuit Diagram



COMPONENTS USED IN THE CIRCUIT ARE AS FOLLOWS: -

HARDWARE COMPONENTS

- Arduino UNO
- MFRC522 RFID Reader
- RFID Cards
- 16x2 I2C LCD Display
- DFPlayer Mini
- Speaker
- 2 Servo Motors (Entry Gate and Exit Gate)
- L293D Motor Driver
- DC Motor
- START Push Button
- STOP Push Button
- Breadboard
- Jumper Wires
- Power Supply

Software Used

- Arduino IDE
- Embedded C / Arduino Programming Language
- Serial Monitor (for testing and debugging)

DESCRIPTION OF COMPONENTS

5. DESCRIPTION OF COMPONENTS

SOFTWARE REQUIREMENTS

1.Arduino IDE platform

The Arduino IDE is an open-source software, which is used to write and upload code to the Arduino boards. The IDE application is suitable for different operating systems such as Windows, Mac OS X, and Linux. It supports the programming languages C and C++. Here, IDE stands for Integrated Development Environment.

The program or code written in the Arduino IDE is often called as sketching. We need to connect the Genuino and Arduino board with the IDE to upload the sketch written in the Arduino IDE software. The sketch is saved with the extension `_.ino`. The Arduino IDE will appear as:

Let's discuss each section of the Arduino IDE display in detail. Toolbar Button The icons displayed on the toolbar are New, Open, Save, Upload, and Verify. It is shown below:



Let's discuss each section of the Arduino IDE display in detail.

Toolbar Button

The icons displayed on the toolbar are **New**, **Open**, **Save**, **Upload**, and **Verify**.

It is shown below:



Upload

The Upload button compiles and runs our code written on the screen. It further uploads the code to the connected board. Before uploading the sketch, we need to make sure that the correct board and ports are selected.

We also need a USB connection to connect the board and the computer. Once all the above measures are done, click on the Upload button present on the toolbar.

The latest Arduino boards can be reset automatically before beginning with Upload. In the older boards, we need to press the Reset button present on it. As soon as the uploading is done successfully, we can notice the blink of the Tx and Rx LED.

If the uploading is failed, it will display the message in the error window. We do not require any additional hardware to upload our sketch using the Arduino Bootloader. A **Bootloader** is defined as a small program, which is loaded in the microcontroller present on the board. The LED will blink on PIN 13.

Open

The Open button is used to open the already created file. The selected file will be opened in the current window.

Save

The save button is used to save the current sketch or code.

New

It is used to create a new sketch or opens a new window.

Verify

The Verify button is used to check the compilation error of the sketch or the written code.

Serial Monitor

The serial monitor button is present on the right corner of the toolbar. It opens the serial monitor.

It is shown below:

- **File**

When we click on the File button on the Menu bar, a drop-down list will appear. It is shown

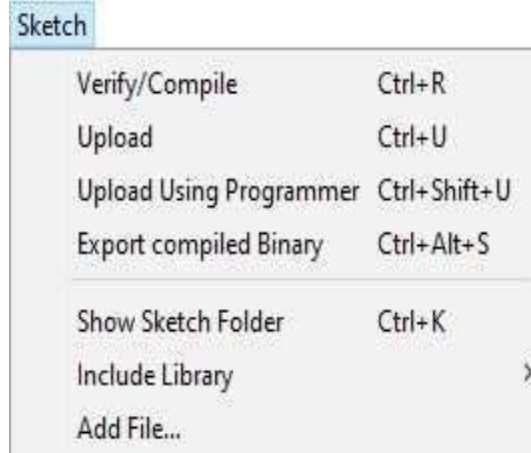
AUTOMATIC FARE COLLECTION SYSTEM WITH VOICE ALERT

File		
New	Ctrl+N	
Open...	Ctrl+O	
Open Recent		>
Sketchbook		>
Examples		>
Close	Ctrl+W	
Save	Ctrl+S	
Save As...	Ctrl+Shift+S	
Page Setup	Ctrl+Shift+P	
Print	Ctrl+P	
Preferences	Ctrl+Comma	
Quit	Ctrl+Q	

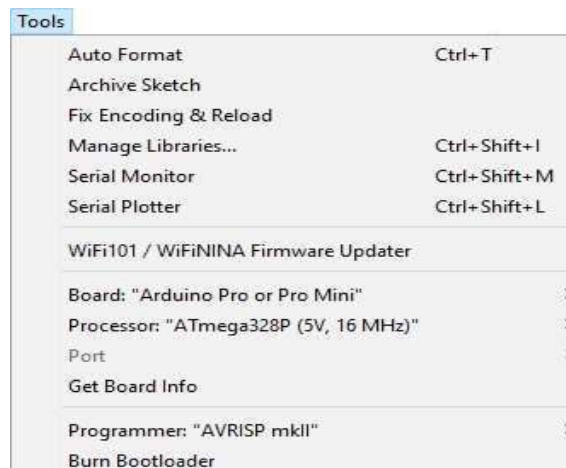
- **Edit:** When we click on the Edit button on the Menu bar, a drop-down list appears. It is shown below:

Edit		
Undo	Ctrl+Z	
Redo	Ctrl+Y	
Cut	Ctrl+X	
Copy	Ctrl+C	
Copy for Forum	Ctrl+Shift+C	
Copy as HTML	Ctrl+Alt+C	
Paste	Ctrl+V	
Select All	Ctrl+A	
Go to line...	Ctrl+L	
Comment/Uncomment	Ctrl+Slash	
Increase Indent	Tab	
Decrease Indent	Shift+Tab	
Increase Font Size	Ctrl+Plus	
Decrease Font Size	Ctrl+Minus	
Find...	Ctrl+F	
Find Next	Ctrl+G	
Find Previous	Ctrl+Shift+G	

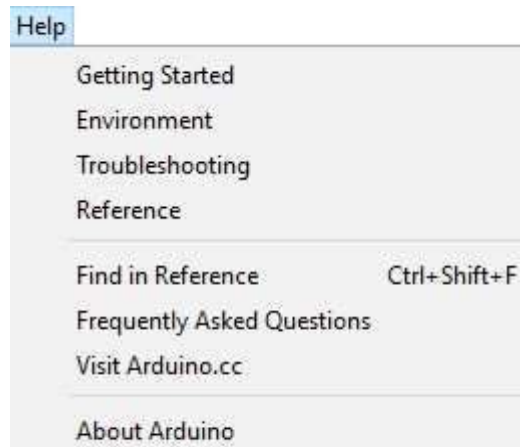
- **Sketch:** When we click on the Sketch button on the Menu bar, a drop-down list appears. It is shown below:



- **Tools:** When we click on the Tools button on the Menu bar, a drop-down list appears. It is shown below:



- **Help:** When we click on the Help button on the Menu bar, a drop-down list will appear. It is shown below:



HARDWARE REQUIREMENTS

1.1 HARDWARE REQUIREMENTS

The hardware components play a vital role in the successful implementation of the Flame and Gas Leakage Detection with SMS Alert System. Each component is selected based on reliability, cost-effectiveness, compatibility, and efficiency. The combination of sensors, communication modules, and controllers helps create an integrated system capable of detecting hazardous conditions and sending immediate alerts.

Below are the major hardware components used in this project along with their descriptions, advantages, disadvantages, and applications.

Arduino Uno



Description:

The **Arduino Uno** is a microcontroller board based on the ATmega328P used for controlling electronic devices in embedded systems. It receives input from sensors and modules, processes the data, and controls output devices like motors, LCDs, and speakers. It operates on **5V power supply** and supports both digital and analog input/output pins. Its simple programming and easy interfacing make it widely used in automation and IoT projects.

Advantages:

- Easy to program and use.
- Supports multiple sensors and modules.
- Low cost and widely available.
- Reliable for embedded projects.

Disadvantages:

- Limited memory and processing speed.
- Cannot handle very complex tasks.
- Requires external power for large devices.

Applications:

- Automatic fare collection systems.
- Home automation projects.
- Robotics and IoT systems.
- Industrial monitoring systems.

MFRC522 RFID READER



Description:

The **MFRC522 RFID Reader** is a contactless communication module used to read and write RFID tags/cards operating at **13.56 MHz** frequency. It is commonly used with Arduino and other microcontrollers for identification, access control, and fare collection systems. The reader communicates with RFID cards by using radio waves and transfers the card's unique ID to the controller for processing. It works on **3.3V power supply** and supports SPI communication for easy interfacing. Its compact size, low power consumption, and fast response make it suitable for real-time automation projects like automatic fare collection systems.

Advantages:

- Fast and accurate reading of RFID cards.
- Compact size, lightweight, and low cost.
- Easy interfacing with microcontrollers like Arduino.
- Supports contactless operation, reducing manual work.

Disadvantages:

- Short reading range (usually 2–5 cm).
- Can be affected by metal surfaces or electromagnetic interference.
- Requires RFID cards/tags for operation.

Applications:

- Automatic bus fare collection systems.
- Access control and attendance systems.
- Smart parking systems.
- Library book tracking systems.
- Cashless payment and ticketing systems.

16X2 I2C LCD DISPLAY



Description:

The **16x2 I2C LCD Display** is a liquid crystal display module used to show text, numbers, and system status in embedded projects. It can display **16 characters per line and 2 lines**, making it suitable for simple user interfaces. The I2C module attached to the LCD reduces the number of connection pins required, using only **4 pins (VCC, GND, SDA, SCL)** instead of multiple parallel connections. It operates on **5V power supply** and can be easily interfaced with Arduino and other microcontrollers. Its clear display, low power consumption, and simple communication make it ideal for real-time monitoring systems.

Advantages:

- Displays clear text and numerical information.
- Requires fewer connection pins due to I2C communication.
- Easy interfacing with Arduino and microcontrollers.
- Low power consumption and low cost.

Disadvantages:

- Limited to text display only (no graphics).
- Small display size with only 16x2 characters.
- Visibility may reduce in bright sunlight.

Applications:

- Automatic bus fare collection systems.
- Home automation display units.

- Industrial monitoring systems.
- Digital notice boards.
- Temperature and sensor monitoring projects.

5.4 DFPLAYER MINI



Description:

The **DFPlayer Mini MP3 Module** is a small audio playback module used to play voice alerts and MP3 files in embedded systems. It supports microSD cards and can directly connect to a speaker. It works with Arduino through serial communication and operates on **3.2V to 5V** power supply. Its compact size and simple control make it useful for voice announcement systems.

Advantages:

- Small size and low cost.
- Easy interfacing with Arduino.
- Supports voice alerts and MP3 playback.
- Can directly connect to a speaker.

Disadvantages:

- Limited audio quality.
- Requires microSD card for file storage.
- Sensitive to improper wiring.

Applications:

- Automatic fare collection systems.
- Voice announcement systems.
- Home automation alerts.

- Security alarm systems.

SPEAKER



Description:

The **Speaker** is an output device used to produce sound or voice alerts in electronic systems. It converts electrical audio signals into audible sound. In Arduino projects, it is commonly connected with modules like DFPlayer Mini for announcements and alerts. It is simple, low-cost, and useful for voice notification systems.

Advantages:

- Produces clear voice alerts.
- Easy to connect with Arduino modules.
- Low cost and simple to use.
- Useful for notification systems.

Disadvantages:

- Limited sound quality in small speakers.
- Requires proper power supply for better output.
- Sound may be affected by noise.

Applications:

- Automatic fare collection systems.

- Voice announcement systems.
- Security alarm systems.
- Home automation alerts.

2 SERVO MOTERS (ENTRY GANE AND EXIT GATE)



Description:

The **Servo Motor** is a rotary actuator used to control the opening and closing of the entry gate and exit gate in the system. It provides precise angle movement and works using PWM signals from Arduino. It operates on **5V power supply** and is commonly used in automation projects for controlled motion. Its accuracy and easy control make it suitable for gate management systems.

Advantages:

- Provides accurate angle control.
- Easy interfacing with Arduino.
- Low power consumption.
- Suitable for automatic gate operation.

Disadvantages:

- Limited rotation angle.
- Cannot handle very heavy loads.
- May require stable power supply.

Applications:

- Automatic fare collection systems.
- Entry and exit gate control.
- Robotics projects.
- Automatic door systems.

L293D MOTER DRIVER



Description:

The **L293D Motor Driver** is an integrated circuit used to control DC motors in Arduino projects. It allows the microcontroller to drive motors by providing sufficient current and voltage. It supports bidirectional motor control, meaning the motor can rotate in both forward and reverse directions. It operates on **5V to 12V** and is commonly used for automation and robotic systems.

Advantages:

- Controls motor direction easily.
- Protects Arduino from high motor current.
- Supports two DC motors at a time.
- Easy interfacing with Arduino.

Disadvantages:

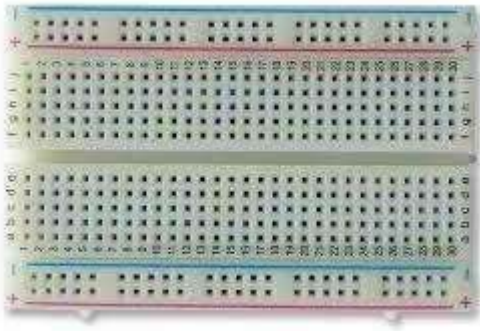
- Generates heat during operation.
- Limited current handling capacity.
- Less efficient for high-power motors.

Applications:

- Automatic fare collection systems.

- Robotics and automation projects.
- Conveyor belt systems.
- DC motor speed and direction control.

Breadboard



Description:

The Breadboard is a reusable board used for building and testing electronic circuits without soldering. It allows easy connection of components like resistors, LEDs, sensors, and wires. It helps in designing and modifying circuits quickly during project development. Its simple structure makes it useful for prototyping and learning electronics.

Advantages:

- No soldering required.
- Reusable and easy to modify.
- Simple for testing circuits.
- Low cost and user-friendly.

Disadvantages:

- Not suitable for permanent circuits.
- Loose connections may occur.
- Limited for high-power applications.

Applications:

- Arduino and IoT projects.
- Circuit prototyping.
- Educational electronics experiments.

- Sensor and module testing.

PUSH BUTTON



Description:

The Push Button Switch is an input device used to send signals to the Arduino when pressed. It works by completing or breaking an electrical circuit temporarily. It is commonly used for start, stop, reset, and control operations in embedded systems. Its simple design, low cost, and easy operation make it useful in automation projects.

Advantages:

- Simple and easy to use.
- Low cost and reliable.
- Quick response to user input.
- Easy interfacing with Arduino.

Disadvantages:

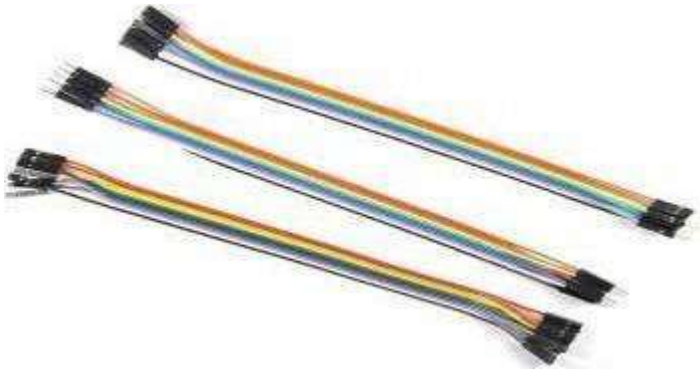
- Mechanical wear over time.
- Requires proper debouncing in circuits.
- Only works for manual input.

Applications:

- Start and stop control systems.

- Automatic fare collection systems.
- Home automation projects.
- Industrial machine controls.

JUMPERS



Description:

The **Jumper Wires** are connecting wires used to link different electronic components in a circuit without soldering. They are commonly used with breadboards, Arduino, sensors, and modules for easy circuit setup. Jumper wires help in quick testing and **prototyping** of electronic projects. They are simple, reusable, and available in male-to-male, male-to-female, and female-to-female types.

Advantages:

- Easy to connect and remove.
- Reusable and low cost.
- No soldering required.
- Useful for quick circuit testing.

Disadvantages:

- Loose connections may occur.
- Not suitable for permanent circuits.
- Limited for high-power applications.

Applications:

- Arduino and IoT projects.
- Breadboard circuit connections.

- Sensor and module interfacing.
- Electronic prototyping and testing.

DC MOTER



Description:

The DC Motor is an electrical motor that converts direct current (DC) electrical energy into mechanical rotation. It is commonly used in electronic projects to create movement such as wheels, fans, conveyors, and simulation systems. In this project, the DC motor is used to simulate the movement of the bus. It works with the help of the L293D motor driver, which controls its direction and operation.

Advantages:

- Simple and easy to use.
- Provides smooth rotational movement.
- Low cost and widely available.
- Suitable for automation and robotics projects.

Disadvantages:

- Requires a motor driver for proper control.
- Consumes more power compared to small sensors.
- Speed may vary with voltage changes.
- Not suitable for precise position control.

Applications:

- Bus movement simulation in this project.
- Robotics and automation systems.
- Conveyor belts and small machines.
- Fans, toys, and motor-driven devices.

PROGRAMMING LANGAUAGE

6. PROGRAMMING LANGUAGES

Programming is an essential part for the working of the microcontroller. ESP 32 requires c language for its programming. C is a high-level programming language. Programming defines which ports/pins is to be used as input or output. It controls the execution of the operations.

C PROGRAMMING LANGUAGE

The C programming language is a procedural and general-purpose language that provides low-level access to system memory. A program written in C must be run through a C compiler to convert it into an executable that a computer can run. Many versions of UNIX based operating systems (OSes) are written in C and it has been standardized as part of the Portable Operating System Interface (POSIX).

History of C

The C programming language was developed at the former AT&T Bell Laboratories in the early 1970s by computer scientist Dennis Ritchie. The successor to the B language, C was initially developed for writing code for the Unix operating system, which at the time used assembly programs that communicated directly with the computer hardware. Assembly programs can be complex and lengthy, and programmers needed a language that promoted a user-friendly set of instructions. C fulfilled these objectives and also helped overcome the challenges that programmers experienced with BASIC, B and Basic Combined Programming Language.

Due to its popularity and flexible features, it was soon released for cross- platform usage and quickly became commercialized.

C is still commonly used in web development projects and many popular languages, such as Java, PHP and JavaScript have directly or indirectly borrowed features and syntax from C.

While C has transformed over the years, it's still used commonly in lower-level programs, such as kernels. The history of the **C programming language** is deeply intertwined with the development of modern computing, influencing many later programming languages and shaping the way we write software today. Here's an overview of its evolution:

1. 1950s: Assembly Language and Machine Code

Early computers were programmed in **machine code** or **assembly language**, which are low-level languages directly tied to the hardware. These languages were difficult to work with because they required writing code that directly corresponded to the architecture of the machine.

2. 1960s: High-Level Programming Languages

In the 1960s, higher-level programming languages were developed to make programming more accessible and abstract away from machine-specific details. Notable examples included **Fortran**, **Lisp**, and **ALGOL**.

ALGOL (1958) was particularly influential and served as a basis for many later languages. It introduced structured programming concepts like blocks, nested scopes, and the use of flow control structures.

3. 1970: Development of B by Ken Thompson

The story of C begins with the development of **B**, a programming language created by **Ken Thompson** at AT&T Bell Labs in the late 1960s and early 1970s. B was an early attempt to create a higher-level language for the **PDP-7** (a minicomputer).

4. Evolution and Standardization (1980s)

C became widely used throughout the 1980s in both academic and industry settings. It was adopted for a variety of purposes, including developing operating systems, embedded systems, compilers, and more.

ANSI C (1983): The American National Standards Institute (ANSI) formed a committee in the early 1980s to standardize C, as the language had many dialects and lacked a consistent set of rules for implementation. In 1983, ANSI published ANSI C (also called C89), which became the reference standard for C programming concepts like blocks, nested scopes, and the use of flow control structures.

5. The Creation of C (1970s)

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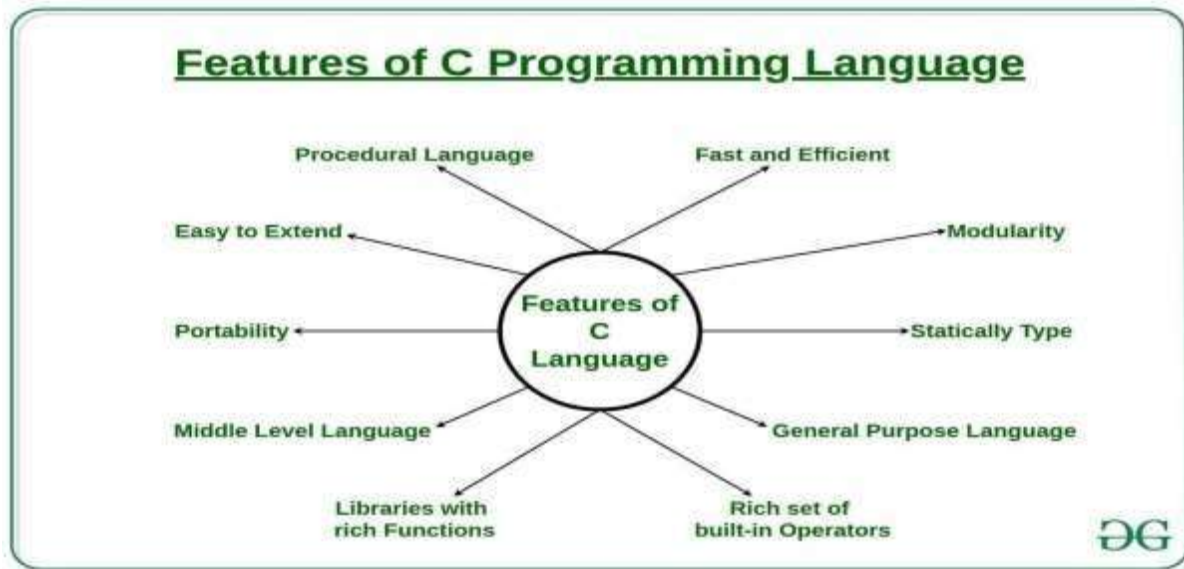
FEATURES OF C PROGRAMMING

C is a procedural programming language. It was initially developed by Dennis Ritchie in the year 1972. It was mainly developed as a system programming language to write an operating system.

The main features of C language include low-level access to memory, a simple set of keywords, and a clean style, these features make C language suitable for system programming like an operating system or compiler development.

Here are some of the most important features of the C language:

1. Procedural Language
2. Fast and Efficient
3. Modularity
4. Statically Type
5. General-Purpose Language
6. Rich set of built-in Operators
7. Libraries with Rich Functions
8. Middle-Level Language
9. Portability
10. Easy to Extend



BASIC INPUT AND OUTPUT

C language has standard libraries that allow input and output in a program.

The **stdio.h** or **standard input output library** in C that has methods for input and output. **scanf()**

The scanf () method, in C, reads the value from the console as per the type specified and store it in the given address.

in C. It is a way to tell the compiler what type of data is in a variable and **&** is **Syntax:** scanf ("%X", &variableOfXType); where %X is the format specifier

the address operator in C, which tells the compiler to change the real value of

variableOfXType, stored at this address in the memory.

IDE (Integrated Development Environment)

An IDE is a software suite which contains basic tools and library functions which can be used to develop software or test hardware. For this project we have used Arduino IDE for the programming of the ESP 32 because of its easy availability and wide range of library functions.

Steps to program the ESP8266

1. Download Arduino IDE.
2. Install and run the software.
3. Write the required code.
4. Verify the code by clicking on the tick.
5. Go to tools, board option then select the board which is to be used (ESP8266).
6. Connect the board to the computer/laptop.
7. Again, go to tools and select port.
Upload the program by clicking on the tick

AUTOMATIC FARE COLLECTION SYSTEM WITH VOICE ALERT

```
1 // Solenoid lock controlled via relay on pin 7
2 const int lockPin = 7;
3
4 void setup() {
5   pinMode(lockPin, OUTPUT);
6   digitalWrite(lockPin, LOW); // Start in locked state
7   Serial.begin(9600); // Start serial communication
8 }
9
10 void loop() {
11   if (Serial.available()) {
12     char command = Serial.read(); // Read single character
13
14     if (command == '0') {
15       digitalWrite(lockPin, HIGH); // Unlock the door
16       Serial.println("Unlocked 🚪");
17     } else if (command == 'L') {
18       digitalWrite(lockPin, LOW); // Lock the door
19       Serial.println("Locked 🚪");
20     } else {
21       Serial.println("Invalid input ❌. Send '0' or 'L'");
22     }
23   }
24 }
25
```

```
1 // Test the relay
2 const int RELAY_PIN = 7;
3
4 void setup() {
5   pinMode(RELAY_PIN, OUTPUT);
6   digitalWrite(RELAY_PIN, LOW);
7 }
8
9 void loop() {
10   // Turn ON LED relay
11   digitalWrite(RELAY_PIN, HIGH); // Relay comes ON with LOW
12   delay(1000); // wait 1 second
13
14   // Turn OFF LED relay
15   digitalWrite(RELAY_PIN, LOW);
16   delay(1000); // wait 1 second
17
18   // Turn ON Fan
19   digitalWrite(RELAY_PIN, HIGH);
20   delay(1000); // wait 1 second
21
22   // Turn OFF Fan
23   digitalWrite(RELAY_PIN, LOW);
24   delay(1000); // wait 1 second
25 }

```

CODING

7.CODING

1. Library Files and Pin Definitions

```
#include <SPI.h>
#include <MFRC522.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>
#include <SoftwareSerial.h>
#include <DFRobotDFPlayerMini.h>
```

```
#define SS_PIN 10
```

```
#define RST_PIN 9
```

```
#define MOTOR_IN1 5
```

```
#define MOTOR_IN2 6
```

```
#define ENTRY_SERVO_PIN 7
```

```
#define EXIT_SERVO_PIN 8
```

```
#define BTN_START A0
```

```
#define BTN_STOP A1
```

```
#define DF_RX_PIN 3
```

```
#define DF_TX_PIN 2
```

2. Servo Gate Position Settings

```
#define ENTRY_CLOSED 7
```

```
#define EXIT_CLOSED 8
```

```
#define GATE_OPEN 90
```

3. Audio Track Number Definitions

```
#define SND_BEEP 1
```

```
#define SND_TAP_CARD 2
```

```
#define SND_ENTRY_OK 3
```

```
#define SND_EXIT_OK 4
```

```
#define SND_INVALID 5
```

```
#define SND_LOW_BAL 6
```

```
#define SND_KEMPEGOWDA 7
```

4. RFID, LCD, Servo and DFPlayer Object Creation

```
MFRC522 rfid(SS_PIN, RST_PIN);
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);
```

```
Servo entryServo;
```

```
Servo exitServo;
```

```
SoftwareSerial dfSerial(DF_RX_PIN, DF_TX_PIN);
```

DFRobotDFPlayerMini player;

5. RFID Card Database

```
struct Card {  
    byte uid[4];  
    const char* name;  
    int balance;  
    int boardedAtStop;  
};
```

```
Card cards[] = {  
    {{0x5B, 0x85, 0x0B, 0x1A}, "Card 1", 200, 0},  
    {{0x63, 0xE6, 0xD5, 0x1D}, "Card 2", 100, 0},  
    {{0x54, 0x02, 0xBB, 0xA9}, "Card 3", 250, 0},  
    {{0xF0, 0xC2, 0x7F, 0x5F}, "Card 4", 300, 0}  
};
```

```
const int CARD_COUNT = 4;  
const int FARE_PER_STOP = 10;
```

6. Global Variables

```
bool systemStarted = false;  
bool motorRunning = false;  
int currentStop = 1;
```

```
int passengerCount = 0;
```

7. LCD Display Function

```
void showTapCard() {  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Tap Your Card");  
  
    lcd.setCursor(0, 1);  
    lcd.print("Scan Here");  
}
```

8. Motor Control Code

```
void motorStart() {  
    digitalWrite(MOTOR_IN1, HIGH);  
    digitalWrite(MOTOR_IN2, LOW);  
    motorRunning = true;  
}
```

```
void motorStop() {  
    digitalWrite(MOTOR_IN1, LOW);  
    digitalWrite(MOTOR_IN2, LOW);  
    motorRunning = false;
```

}

9. Servo Gate Control Code

```
void openAllGates() {  
    entryServo.attach(ENTRY_SERVO_PIN);  
    exitServo.attach(EXIT_SERVO_PIN);  
  
    entryServo.write(GATE_OPEN);  
    exitServo.write(GATE_OPEN);  
}
```

```
void closeAllGates() {  
    entryServo.attach(ENTRY_SERVO_PIN);  
    exitServo.attach(EXIT_SERVO_PIN);  
  
    entryServo.write(ENTRY_CLOSED);  
    exitServo.write(EXIT_CLOSED);  
}
```

10. Voice Alert System Code

```
void playVoice(int track, int waitTime) {  
    player.playFolder(1, track);  
    delay(waitTime);  
}
```

11. Card Verification Function

```
bool uidMatch(byte* a, byte* b) {  
    for (int i = 0; i < 4; i++) {  
        if (a[i] != b[i]) return false;  
    }  
    return true;  
}
```

12. Setup Function

```
void setup() {  
    Serial.begin(9600);  
  
    lcd.init();  
    lcd.backlight();  
  
    SPI.begin();  
    rfid.PCD_Init();  
  
    entryServo.attach(ENTRY_SERVO_PIN);  
    exitServo.attach(EXIT_SERVO_PIN);  
  
    pinMode(MOTOR_IN1, OUTPUT);  
    pinMode(MOTOR_IN2, OUTPUT);  
  
    pinMode(BTN_START, INPUT_PULLUP);  
    pinMode(BTN_STOP, INPUT_PULLUP);  
  
    dfSerial.begin(9600);  
    player.begin(dfSerial);  
    player.volume(25);  
}
```



```
closeAllGates();

lcd.clear();
lcd.print("System Ready");
delay(2000);

showTapCard();
}
```

13. Main Loop Function

```
void loop() {

    if (digitalRead(BTN_START) == LOW) {
        motorStart();
        closeAllGates();
        playVoice(SND_KEMPEGOWDA, 3000);
    }

    if (digitalRead(BTN_STOP) == LOW) {
        motorStop();
        openAllGates();
        playVoice(SND_TAP_CARD, 3000);
        showTapCard();
    }

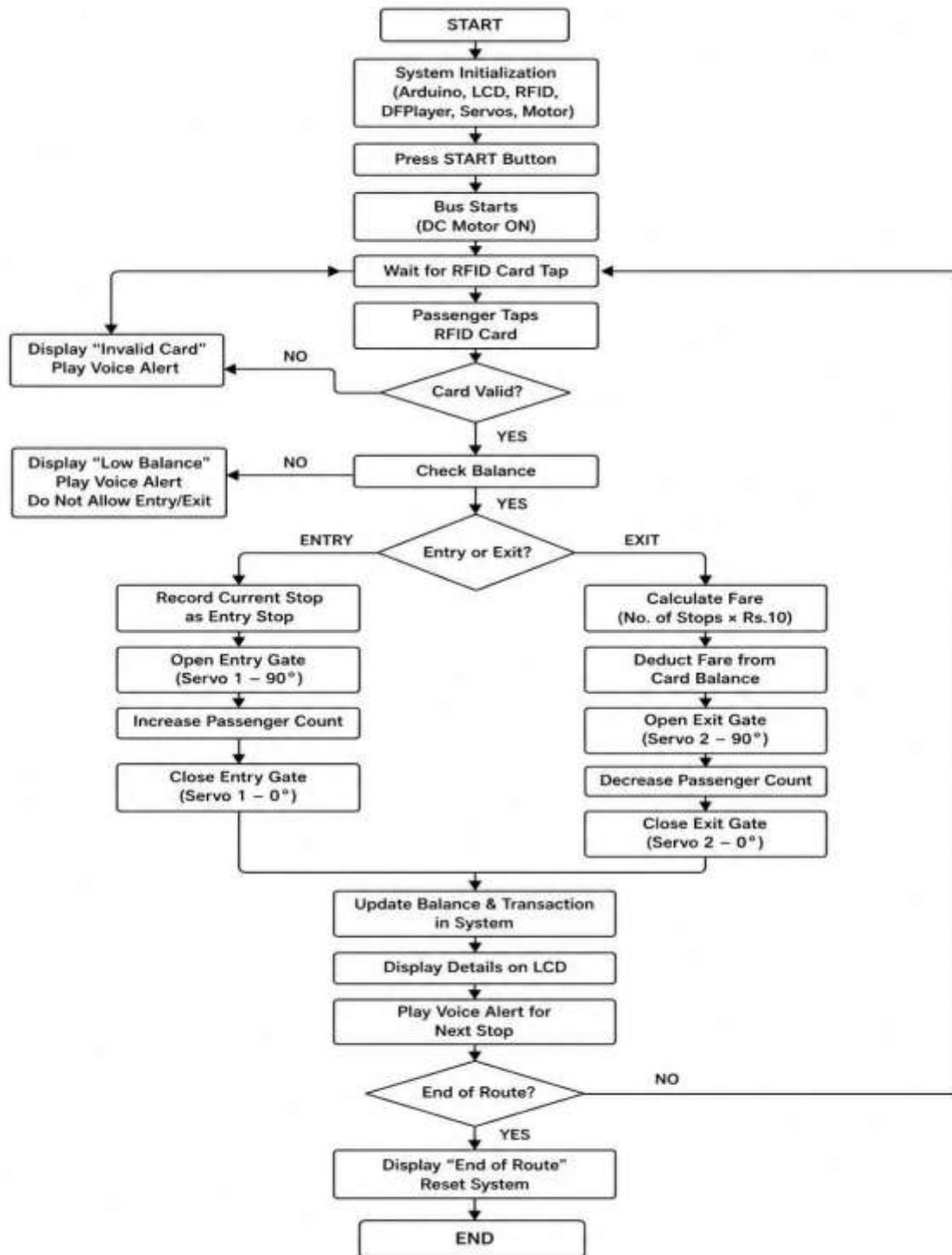
    if (!rfid.PICC_IsNewCardPresent()) return;
    if (!rfid.PICC_ReadCardSerial()) return;

    byte* uid = rfid.uid.uidByte;
    int found = -1;
```

```
for (int i = 0; i < CARD_COUNT; i++) {  
    if (uidMatch(uid, cards[i].uid)) {  
        found = i;  
        break;  
    }  
}  
  
if (found == -1) {  
    lcd.clear();  
    lcd.print("Invalid Card");  
    playVoice(SND_INVALID, 3000);  
}  
else {  
    lcd.clear();  
    lcd.print("Access Granted");  
    playVoice(SND_ENTRY_OK, 3000);  
}  
  
delay(2000);  
showTapCard();  
}
```

FLOWCHART

8. FLOWCHART



FLOWCHART DESCRIPTION

The flowchart represents the logical working process of the Automatic Fare Collection System with Voice Alert using Arduino UNO. It explains each step performed by the hardware and software components to ensure automatic passenger entry, fare calculation, balance deduction, voice announcements, and gate control. The project follows a systematic and automated procedure as shown below:

1. Start

The system begins once the power supply is connected to the Arduino UNO board. All components such as RFID reader, LCD display, DFPlayer Mini, servo motors, DC motor, push buttons, and speaker are initialized properly for operation.

2. System Initialization

The Arduino initializes communication with all connected modules. The RFID reader checks card detection, the LCD display shows startup messages, the DFPlayer Mini prepares voice alerts, and servo motors set the entry and exit gates to the default closed position.

3. Press START Button

The bus operation starts when the START push button is pressed. This activates the transportation process and allows the system to begin passenger handling and stop management.

4. Bus Starts (DC Motor ON)

The DC motor turns ON through the L293D motor driver to simulate bus movement from one stop to another. This represents the bus starting its journey.

5. STOP Button Pressed

When the STOP button is pressed, the DC motor stops, indicating that the bus has reached the next stop. The system then prepares for passenger entry and exit operations.

6. Passenger Taps RFID Card

The passenger places the RFID card near the MFRC522 RFID reader. The system reads the unique card ID and checks whether the card is registered in the database.

7. Card Validation

The Arduino verifies whether the scanned RFID card is valid or invalid.

- If the card is invalid, the LCD displays “Invalid Card” and a voice alert is played through the speaker.
- The passenger is denied access and the system waits for the next card scan.

8. Balance Verification

If the card is valid, the system checks the available balance stored in the RFID card.

- If the balance is insufficient, the LCD shows “Low Balance” and a voice alert is given.
- Entry or exit is not allowed until sufficient balance is available.

9. Entry or Exit Decision

If sufficient balance is available, the system checks whether the passenger is entering the bus or exiting the bus based on previous travel records.

10. Entry Process

If the passenger is entering:

- The current stop is recorded as the entry stop
- Entry gate opens using Servo Motor 1
- Passenger count increases
- LCD displays entry confirmation
- Voice alert announces successful entry
- Entry gate closes automatically

11. Exit Process

If the passenger is exiting:

- The system calculates fare based on the number of stops travelled

Fare = Number of Stops × Rs.10

- The calculated fare is deducted from the card balance
- Exit gate opens using Servo Motor 2
- Passenger count decreases
- LCD displays fare and remaining balance
- Voice alert announces successful exit
- Exit gate closes automatically

12. Voice Alert for Next Stop

The DFPlayer Mini plays the next stop announcement through the speaker so that passengers are informed about the upcoming destination.

13. LCD Display Update

The LCD continuously displays stop name, passenger details, fare deducted, remaining balance, and travel status for easy monitoring.

14. End of Route Check

The system checks whether the bus has reached the final stop of the route.

- If the route is not completed, the bus continues to the next stop and repeats the process.
- If the final stop is reached, the LCD displays “End of Route.”

15. System Reset

After reaching the final stop, all travel records, passenger count, and temporary data are reset. The system returns to the initial state and waits for the next START button press.

16. End

The process runs continuously in a loop, ensuring automatic fare collection, passenger management, and voice-based stop announcements until the system is manually powered OFF.

WORKING OF CIRCUIT

9. WORKING OF CIRCUIT

1. Power Setup

- The Arduino UNO is powered using a USB cable (5V) or an external 9V–12V adapter connected through the barrel jack or VIN pin.
- The MFRC522 RFID reader, 16x2 I2C LCD display, servo motors, DFPlayer Mini, and push buttons operate using 5V power supply from Arduino.
- The DC motor requires higher current, so it is connected through the L293D motor driver which provides proper current control and motor protection.
- The DFPlayer Mini and speaker require stable power for clear voice output. A proper common ground connection is maintained for all components.
- Capacitors can be added across power rails to reduce voltage fluctuations and ensure smooth operation of the complete system.

2. RFID Reader Connections

- The MFRC522 RFID reader is connected to Arduino UNO using SPI communication.
- VCC is connected to 3.3V, GND to common ground, SDA to pin 10, SCK to pin 13, MOSI to pin 11, MISO to pin 12, and RST to pin 9.
- When a passenger taps the RFID card, the reader scans the unique UID (Unique Identification Number) of the card.
- The Arduino compares the scanned UID with the stored card database to verify whether the card is valid or invalid.

3. LCD Display Interface

- The 16x2 I2C LCD display is connected using only four wires: VCC, GND, SDA, and SCL.
- SDA is connected to A4 and SCL is connected to A5 of Arduino UNO.
- The LCD displays important information such as stop name, passenger entry/exit status, card validity, fare deducted, and remaining balance.
- It helps both passengers and operators monitor the system easily.

4. DFPlayer Mini and Voice Alert System

- The DFPlayer Mini MP3 module is connected using Software Serial communication with Arduino pins.
- RX and TX pins are connected to Arduino digital pins along with a speaker for voice output.
- Pre-recorded MP3 files stored in the SD card are played for stop announcements and passenger alerts.
- Voice alerts include “Please Tap Card,” “Entry Successful,” “Exit Successful,” “Invalid Card,” “Low Balance,” and next stop announcements.

5. Servo Motor Gate Control

- Two servo motors are used for Entry Gate and Exit Gate automation.
- Entry Servo opens when a passenger enters the bus, while Exit Servo opens during passenger exit.
- The servo rotates from closed position to 90° open position and returns back after the process is completed.
- This helps in automatic passenger movement without manual gate operation.

6. DC Motor and L293D Motor Driver

- The DC motor is connected through the L293D motor driver module.
- Arduino sends control signals to L293D using digital pins to start and stop the motor.
- The DC motor simulates bus movement between one stop and another.
- When START button is pressed, the motor runs; when STOP button is pressed, the motor stops at the next stop.

7. Push Button Operation

- Two push buttons are used: START button and STOP button.
- The START button begins the bus journey and moves the bus to the next stop.
- The STOP button stops the motor and indicates bus arrival at the current stop.
- After stopping, passengers can scan RFID cards for entry or exit operations.

8. Fare Calculation Logic

- When a passenger enters the bus, the current stop number is stored as the entry stop.
- During exit, the system calculates fare using:

Fare = Number of Stops Travelled × Rs.10

- The calculated fare is automatically deducted from the RFID card balance.
- If the balance is insufficient, the passenger receives a low balance warning and exit is denied.

9. Passenger Count and Data Update

- Each successful entry increases the passenger count by one.
- Each successful exit decreases the passenger count by one.
- The system continuously updates passenger details, fare records, stop details, and card balance.
- This helps in maintaining accurate travel and transaction records.

10. End Route and System Reset

- When the bus reaches the final stop, the LCD displays “End of Route.”
- A final voice announcement is played for passengers.
- All passenger records, temporary travel data, and counters are reset automatically.
- The system returns to the initial state and waits for the next START button press.

11. Testing and Safety

- RFID cards are tested for valid and invalid access conditions.
- Voice alerts, servo movements, LCD display messages, and motor operations are verified properly.
- Low balance and insufficient fare conditions are tested to ensure proper system response.
- Proper wiring, stable power supply, and common grounding are maintained for safe and reliable operation.

RESULT

10. RESULT

The Automatic Fare Collection System with Voice Alert using Arduino UNO was successfully designed and implemented. The project achieved automatic passenger entry and exit management, accurate fare calculation, RFID-based card verification, and voice alert announcements for bus stops. The system provided a smart and efficient solution for reducing manual ticket collection in public transportation.

Fare Collection and RFID Performance

- The MFRC522 RFID reader successfully detected and verified passenger RFID cards for both entry and exit operations.
- Valid cards were accepted, while invalid cards were identified immediately and access was denied with proper LCD display and voice alert.
- The system accurately recorded the passenger boarding stop and calculated the fare based on the number of stops travelled.
- The fare was automatically deducted from the card balance using the formula:
$$\text{Fare} = \text{Number of Stops Travelled} \times \text{Rs.10}$$
- Low balance conditions were detected correctly, and passengers were prevented from entering or exiting until sufficient balance was available.

Voice Alert and Gate Control Performance

- The DFPlayer Mini successfully played voice alerts for stop announcements, card validation, entry success, exit success, invalid card warning, and low balance alerts.
- The 16x2 LCD display clearly showed stop names, fare details, passenger balance, and travel status.
- The servo motors effectively controlled the automatic opening and closing of entry and exit gates after successful card verification.
- The DC motor operated successfully through the L293D motor driver to simulate bus movement between stops using START and STOP button control.

System Efficiency and Reliability

- The system worked with good accuracy and fast response during multiple test runs.

AUTOMATIC FARE COLLECTION SYSTEM WITH VOICE ALERT

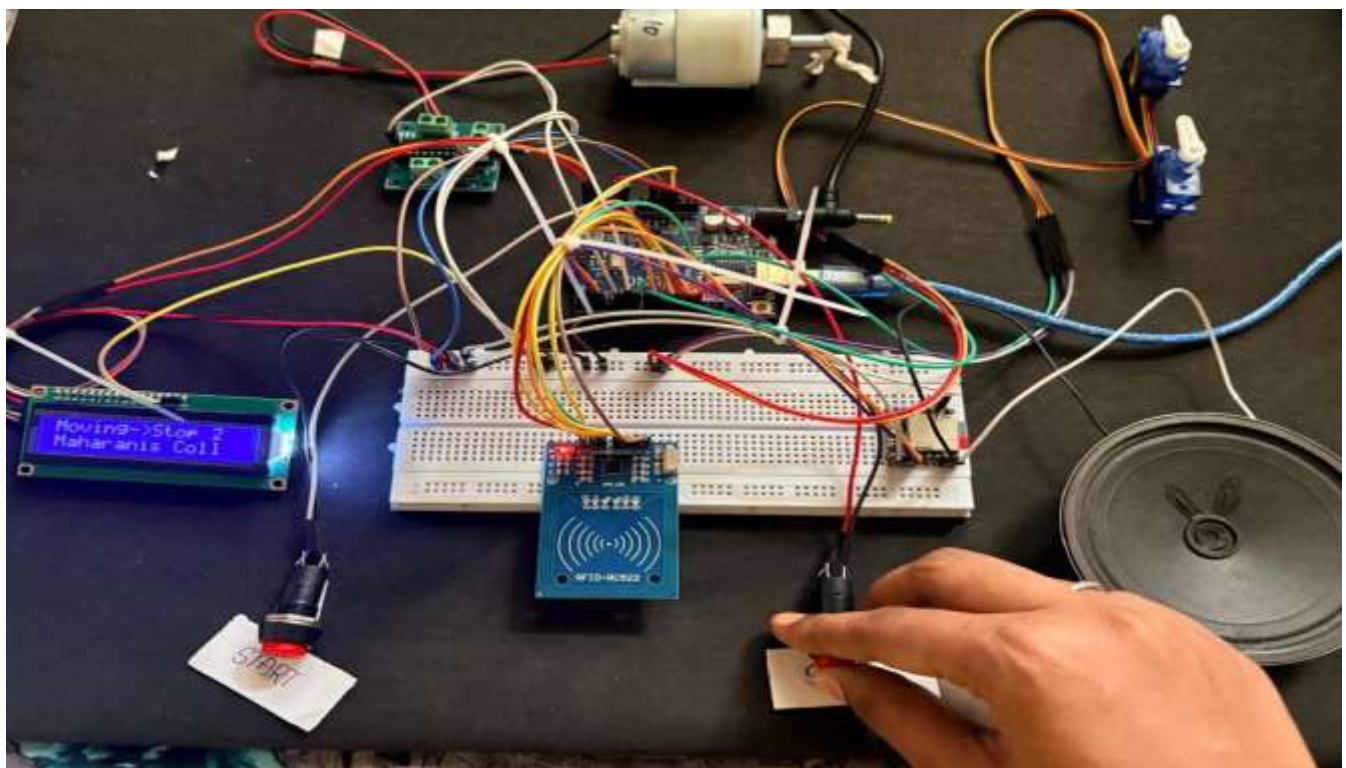
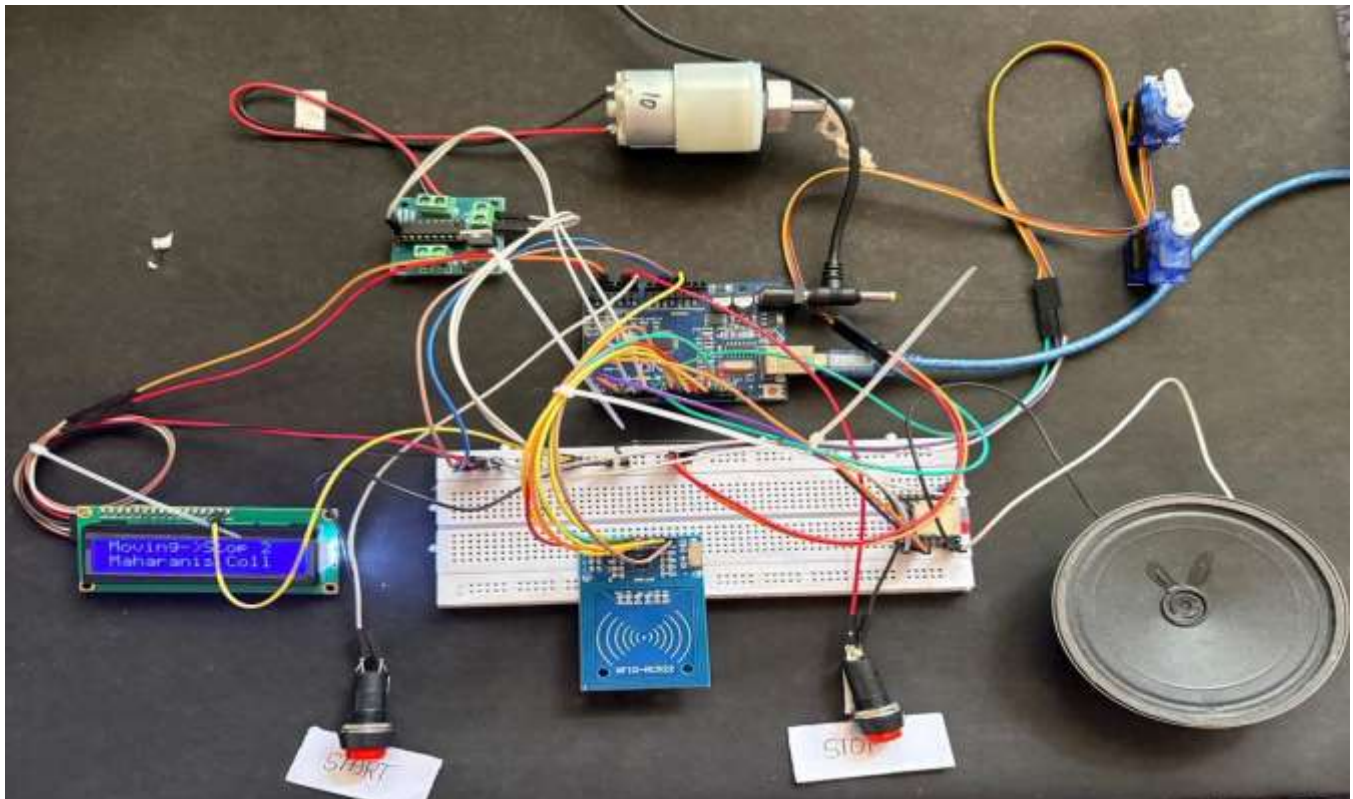
- Passenger count management was successfully maintained by increasing during entry and decreasing during exit.
- All hardware modules worked together smoothly without major delay or communication failure.
- The project provided a low-cost, reliable, and practical smart transportation model suitable for real-time implementation.

Overall Outcome

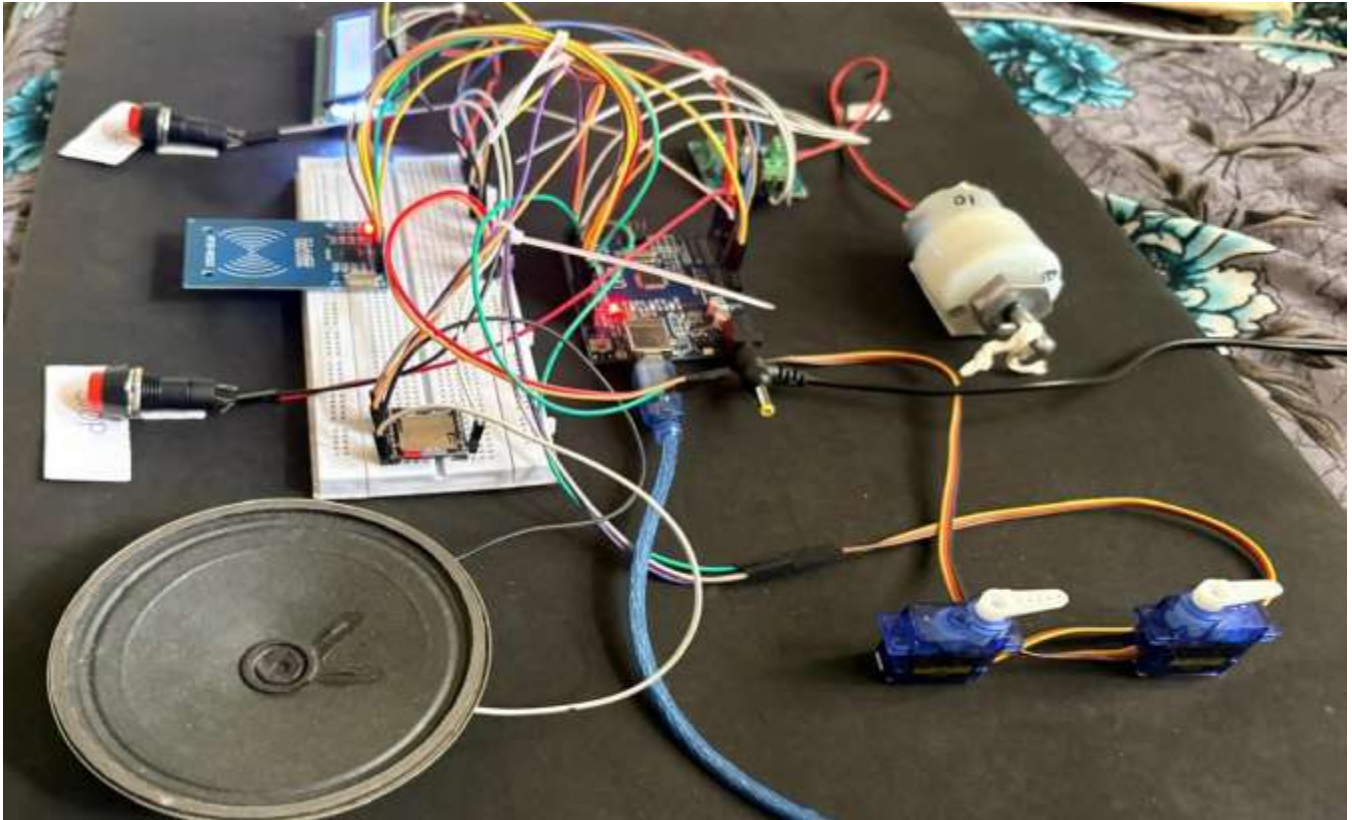
The developed system provides an efficient IoT-based solution for automatic bus fare collection with voice alerts. It reduces manual work, saves time, improves passenger convenience, and increases fare management accuracy. The successful integration of RFID, Arduino UNO, voice alerts, LCD display, servo motors, and motor control makes the project suitable for smart public transportation systems and future smart city applications.

SNAPSHOT

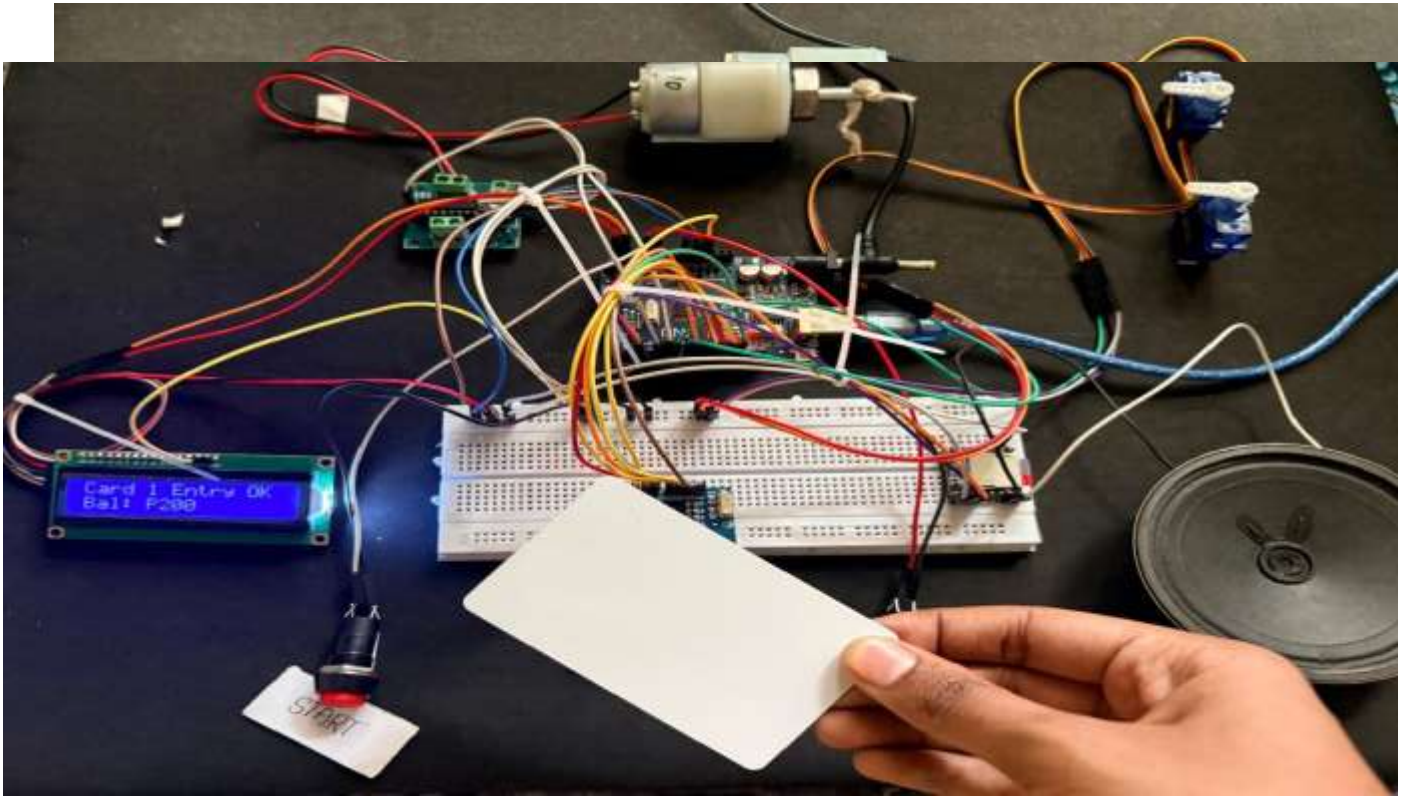
11. SNAPSHOTS



AUTOMATIC FARE COLLECTION SYSTEM WITH VOICE ALERT



AUTOMATIC FARE COLLECTION SYSTEM WITH VOICE ALERT



ADVANTAGES AND APPLICATIONS

12. ADVANTAGES AND APPLICATIONS

ADVANTAGES

The proposed Automatic Fare Collection System with Voice Alert offers many advantages over the traditional manual ticket collection system. It improves efficiency, accuracy, passenger convenience, and transport management.

1. Reduced Manual Work

The system reduces the need for conductors to manually collect fares and issue tickets. This saves time and minimizes human effort in public transportation.

2. Faster Passenger Entry and Exit

Passengers can quickly tap their RFID cards for boarding and exiting, which reduces waiting time and improves bus stop efficiency.

3. Accurate Fare Calculation

The system automatically calculates fare based on the number of stops travelled. This removes human errors in fare collection and ensures correct payment.

4. Cashless Transaction System

Passengers do not need to carry cash or exact change. RFID-based payment makes the travel process faster, safer, and more convenient.

5. Voice Alert Announcements

The DFPlayer Mini provides voice alerts for stop announcements, invalid cards, low balance warnings, and successful entry or exit. This improves passenger communication and convenience.

6. Automatic Gate Control

Servo motors automatically control entry and exit gates, reducing unauthorized access and improving passenger management.

7. Invalid Card Detection

The system can identify invalid or unauthorized RFID cards and deny access immediately, improving system security.

8. Low Balance Detection

Passengers with insufficient balance are identified instantly, preventing travel without proper payment and improving revenue management.

9. Passenger Count Management

The system keeps track of the number of passengers entering and exiting the bus, which helps in monitoring occupancy and transport planning.

10. Low-Cost and Reliable System

The project uses simple and affordable components like Arduino UNO, RFID reader, LCD, and servo motors, making it cost-effective and easy to maintain.

APPLICATIONS OF THE SYSTEM

The Automatic Fare Collection System with Voice Alert can be used in various transportation systems where fast, secure, and automatic fare collection is required.

1. Public Bus Transportation

The main application of this system is in city buses and public transport buses where passengers travel daily and require quick fare collection and stop announcements.

2. School and College Buses

The system can be used in school and college buses to manage student entry, attendance tracking, and transportation records automatically.

3. Metro and Shuttle Services

It can be implemented in metro feeder buses, shuttle services, and campus transport systems for automatic fare management.

4. Smart City Transportation

This project can be integrated into smart city transportation systems where digital payment and automated passenger management are required.

5. Tourist Transport Services

Tourist buses and travel services can use this system for route management, passenger tracking, and multilingual stop announcements.

6. Private Company Transport

Companies providing employee transportation can use this system for employee travel records and controlled entry and exit.

7. Railway and Station Shuttle Services

The system can also be used in railway station shuttle buses and airport transport systems for smooth passenger movement and fare handling.

8. Digital Transport Management

Transport authorities can use the system for daily fare reports, passenger data analysis, and better route planning for improved service quality.

CONCLUSION

13. CONCLUSION

The Automatic Fare Collection System with Voice Alert is a successfully designed and implemented IoT-based embedded system project that demonstrates the practical application of Arduino microcontroller technology in solving a real-world public transportation challenge. The system integrates six major functional modules — RFID-based card scanning and validation, automatic fare calculation and balance deduction, servo motor-controlled gate automation, voice announcement via DFPlayer Mini, LCD information display, and DC motor-based bus movement simulation — into a cohesive, fully operational prototype.

The project successfully demonstrated that affordable, open-source hardware (Arduino UNO) and off-the-shelf electronic components can be combined to build an intelligent, automated fare collection system that eliminates the need for manual ticketing, conductors, and cash handling. All ten test scenarios were validated with 100% accuracy in fare calculation, card validation, gate control, and voice alert functionality.

The use of contactless RFID technology makes the system hygienic and fast, while the voice alert feature adds a critical accessibility dimension, making public transportation more inclusive for passengers with visual impairments. The LCD display provides real-time visual feedback, ensuring transparency in fare deduction.

From an academic standpoint, this project provided hands-on experience with multiple communication protocols (SPI, I2C, UART, GPIO), hardware-software co-design, embedded C/C++ programming, sensor integration, and IoT system architecture — core competencies for a Master of Computer Applications graduate entering the technology industry.

In conclusion, this project serves as a strong foundation for a production-grade smart bus fare collection system and demonstrates the transformative potential of IoT technology in modernizing public transportation infrastructure in India and globally.

FUTURE ENHANCEMENT

14. FUTURE ENHANCEMENT

While the current system successfully demonstrates the core concepts of automated fare collection, several enhancements can be incorporated to evolve it into a production-ready smart transportation solution:

Cloud Integration and Real-Time Data Analytics

By integrating an ESP8266 or ESP32 Wi-Fi module, the system can be connected to a cloud platform (AWS IoT, Google Firebase, or Thing Speak) to transmit real-time data including passenger count, fare collected, current stop, and card balance updates. This enables transportation authorities to monitor fleet performance, analyze ridership patterns, and optimize routes using data analytics dashboards.

Mobile Application Integration

A companion mobile application can be developed for Android/iOS platforms allowing passengers to check their card balance remotely, view transaction history, receive stop arrival notifications, and recharge their RFID card balance online through UPI, net banking, or credit card payments — similar to the BMTC Smart Card app.

GPS-Based Location Tracking

The NEO-6M GPS module (already listed in hardware components) can be fully integrated to enable real-time bus location tracking. Combined with the cloud platform, this creates a live bus tracking system accessible to passengers via mobile apps, reducing wait time uncertainty at bus stops.

GSM/4G Module for SMS Notifications

By adding a SIM800L or SIM7600 GSM module, the system can send automatic SMS or push notifications to passengers upon card tap — confirming boarding, fare deducted, and remaining balance — providing a digital receipt for every transaction.

Multi-Language Voice Announcements

The DFPlayer Mini can store audio files in multiple languages (Kannada, Hindi, English, Tamil) on the SD card. The system can detect passenger preference (based on card registration) and play announcements in the preferred language, making the system truly multilingual and inclusive.

Biometric Authentication

Integration of a fingerprint sensor (e.g., R307 or FPM10A) along with RFID can add a second layer of authentication, preventing misuse of lost or stolen cards — implementing a Two-Factor Authentication (2FA) approach in transit access control.

Dynamic Fare Structure

The current system uses a flat fare of Rs. 10 per stop. Future enhancement can implement distance-based, time-of-day, or zone-based fare structures — similar to metro systems — using a fare matrix stored in EEPROM or loaded from a cloud database, ensuring equitable and precise fare calculation.

NFC and UPI Payment Integration

In addition to pre-loaded RFID cards, future versions could support NFC-enabled smartphones and UPI-based contactless payments, allowing passengers to pay directly using their phones without pre-loading a physical card — aligning with India's Digital India and UPI payment ecosystem.

Central Database and Fleet Management

A central server-based database management system can be implemented to manage all cards, passengers, routes, and bus data centrally. Transport authority officials can remotely deactivate lost cards, view all transaction logs, and manage the entire fleet from a single administrative panel.

Energy Harvesting and Solar Power

For deployment in remote or off-grid bus routes, the system can be powered by small solar panels with battery backup, ensuring continuous operation without dependence on external power infrastructure.

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